TEXORA · LECTURE SHEET

The Carding Machine

Carding is the process that turns opened tufts into individual fibres, removes most of the remaining trash and neps, and delivers the first ordered product, the sliver. Its key calculations — surface speeds, carding draft and production from doffer speed and sliver hank — are worked in full here.

YM-3207 · ADVANCED SPINNING CALCULATIONS · TOPIC 2 · SPECIALIZATION — YARN MANUFACTURING

1. What the card does

The card receives the blowroom lap (about 450 g/m) and must reduce it to a sliver of about 4 g/m while opening it to single fibres. Along the way it removes the fine trash and dust the blowroom could not, disentangles a large share of the neps, blends the material over its working width and length, and begins the parallelisation of fibres that drawing continues. Carding quality is often called the heart of spinning because faults left here — neps, trash particles, uneven sliver — are only partly correctable later.

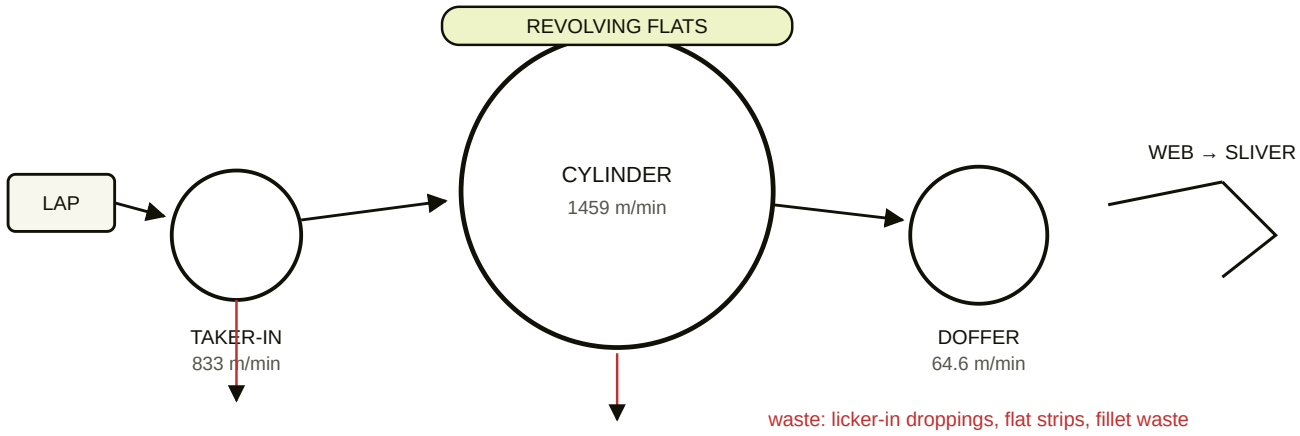


Figure 1 — Working elements of the revolving-flat card with the surface speeds calculated in this sheet. Material flows lap → taker-in → cylinder/flats → doffer → web → sliver.

2. Surface speeds

Carding action depends on the **surface speed** of each clothed element, not merely its rpm, because the elements have very different diameters.

$$\text{Surface speed (m/min)} = \pi \times \text{diameter (m)} \times \text{speed (rpm)}$$

ELEMENT	DIAMETER	SPEED	SURFACE SPEED
Cylinder	1290 mm	360 rpm	π × 1.29 × 360 = **1459 m/min**
Taker-in (licker-in)	241 mm	1100 rpm	π × 0.241 × 1100 = **833 m/min**
Doffer	686 mm	30 rpm	π × 0.686 × 30 = **64.6 m/min**

The enormous speed ratio between cylinder and doffer is the point: fibre carried on the fast cylinder is condensed onto the slow doffer, which is why the doffer's clothing fills and a coherent web can be stripped from it.

3. Carding draft

The draft at the card is the ratio of feed weight to delivered weight per unit length.

$$\text{Draft} = \text{weight of lap per metre} / \text{weight of sliver per metre}$$

WORKED EXAMPLE — DRAFT, SLIVER WEIGHT AND PRODUCTION AT THE CARD

Lap weight = 450 g/m. Delivered sliver hank = 0.15 Ne.

Sliver weight = 590.5 / 0.15 = 3937 tex = **3.94 g/m**.

Draft = 450 / 3.94 = **114.3**.

Production: the doffer (686 mm diameter, 30 rpm) has surface speed 64.6 m/min, and the web leaves at doffer surface speed.

Production = 64.6 m/min × 3.94 g/m = 254.5 g/min = **15.3 kg/h** at 100% efficiency.

At a realistic 85% efficiency: 15.3 × 0.85 = **13.0 kg/h**.

Note how production is fixed by only two quantities — doffer speed and sliver weight. Raising doffer speed raises production but shortens the time fibres spend under the flats, so carding quality per kilogram falls; this trade-off governs every carding decision.

4. Where the card's waste comes from

WASTE	ORIGIN	TYPICAL SHARE OF FEED
Licker-in droppings	Trash and short fibre thrown out under the taker-in grids	2–3%
Flat strips	Short fibre, neps and trash combed out by the flats	2–3%

Cylinder and doffer waste, fillet waste	Fly and stripping waste	0.3–0.5%
Total card waste	—	about 4–6%

Flat strips are the card's cleaning record: a rising strip rate with dirty material is normal, but high strips on clean material mean the flats are taking good fibre — usually a setting or clothing problem. Because roughly 95% of the feed leaves as sliver, the mass balance of Topic 1's blowroom example continues: of 92 kg of lap, about 87 kg becomes sliver.

5. Settings that the calculations assume

The numbers above hold only at correct settings. The critical ones are taker-in to feed plate (controls opening at entry), cylinder to flats — the true carding zone, set in thousandths of an inch (commonly 0.20–0.25 mm) — and cylinder to doffer, which governs transfer. A card whose cylinder–flat setting has opened with wear will still show the calculated production, but its nep removal collapses; production arithmetic and quality control must always be read together (see YM-4101, Topic 4).

Glossary

Carding action — the point-to-point action between cylinder and flat clothing that separates fibre tufts into single fibres. **Taker-in** — the saw-tooth roller that opens the lap and feeds the cylinder. **Flats** — the slowly moving clothed bars over the cylinder where most carding and nep removal occur. **Doffer** — the roller that condenses fibre from the cylinder into a web. **Web** — the thin fibre film stripped from the doffer, gathered into sliver. **Flat strips** — waste stripped from the flats, rich in short fibre and neps. **Hank** — the indirect count of sliver (Ne); higher hank means finer sliver.

Key points

• Surface speed = π × diameter × rpm; cylinder 1459 m/min against doffer 64.6 m/min in the example.
• Carding draft = lap weight/sliver weight per metre = 114.3 for a 450 g/m lap and 0.15 Ne sliver.
• Production = doffer surface speed × sliver weight; 15.3 kg/h at 100%, 13.0 kg/h at 85%.
• Card waste totals about 4–6% of feed, mostly licker-in droppings and flat strips.
• Production figures assume correct settings; a worn cylinder–flat setting keeps output but destroys quality.

Practice

1. A doffer of 686 mm diameter runs at 36 rpm and the sliver delivered is 0.14 Ne. Calculate the production at 100% and at 85% efficiency.
2. A card consumes a lap of 480 g/m and delivers sliver of 4.2 g/m. Calculate the carding draft.
3. Why does doubling doffer speed not double the useful output of a card department? Give two reasons.

Answer key

1. Surface speed = π × 0.686 × 36 = 77.6 m/min. Sliver = 590.5/0.14 = 4218 tex = 4.22 g/m. Production = 77.6 × 4.22 = 327.4 g/min = **19.6 kg/h**; at 85% = **16.7 kg/h**.
2. Draft = 480 / 4.2 = **114.3**.
3. (1) Carding quality falls: fibres spend less time in the cylinder–flat zone, so nep and trash removal drop and sliver evenness worsens — the extra output may fail quality limits. (2) The bottleneck moves: blowroom supply, can handling and downstream drawframes are sized for the old rate, so the department cannot absorb twice the sliver.

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